Project Report

On

Arduino-based Keyboard Emulator

For

Game Controls using Push Buttons

By

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Problem Statement

Classic PC games are intended to be operated with a keyboard or game controller. Nonetheless, for custom arcade-style experiences, particularly in educational or DIY endeavours, there is a demand for straightforward, affordable physical controls that mimic keyboard inputs. This project intends to create an Arduino-powered system that utilizes push buttons to replicate keypresses for directions such as W, A, S, and J, which are often employed in gaming for movements and actions.

Methodology

- **Hardware Setup:**
  - Four push buttons serve as input devices.
  - Every button is linked to a specific digital pin on the Arduino Leonardo.
  - Internal pull-up resistors are activated by using INPUT_PULLUP.

- **Software Logic:**
  - The Arduino constantly observes the status of every button.
  - When a button is activated, it initiates the associated keypress by utilizing the Keyboard library.
  - Upon release, it halts the key press.
  - The HID functionality is initiated by Keyboard.begin().
- **Key Mapping:**
  - Pin 4 → 'W' key (Up)
  - Pin 5 → 'A' key (Left)
  - Pin 6 → 'S' key (Down)
  - Pin 7 → 'J' key (Right or Action)

Schematic

- Button 1 (Up): One terminal to Pin 4, other to GND
- Button 2 (Left): One terminal to Pin 5, other to GND
- Button 3 (Down): One terminal to Pin 6, other to GND

- Button 4 (Right): One terminal to Pin 7, other to GND

Internal pull-up resistors are enabled in the code, so no external resistors are needed.

Components Used

I. **Arduino Leonardo**

Offers built-in USB HID capabilities, enabling it to mimic a keyboard.

II. **Aluminium Sheets (x8)**

Utilized to transmit signals to the Arduino upon being pressed and released

III. **Connecting wires**

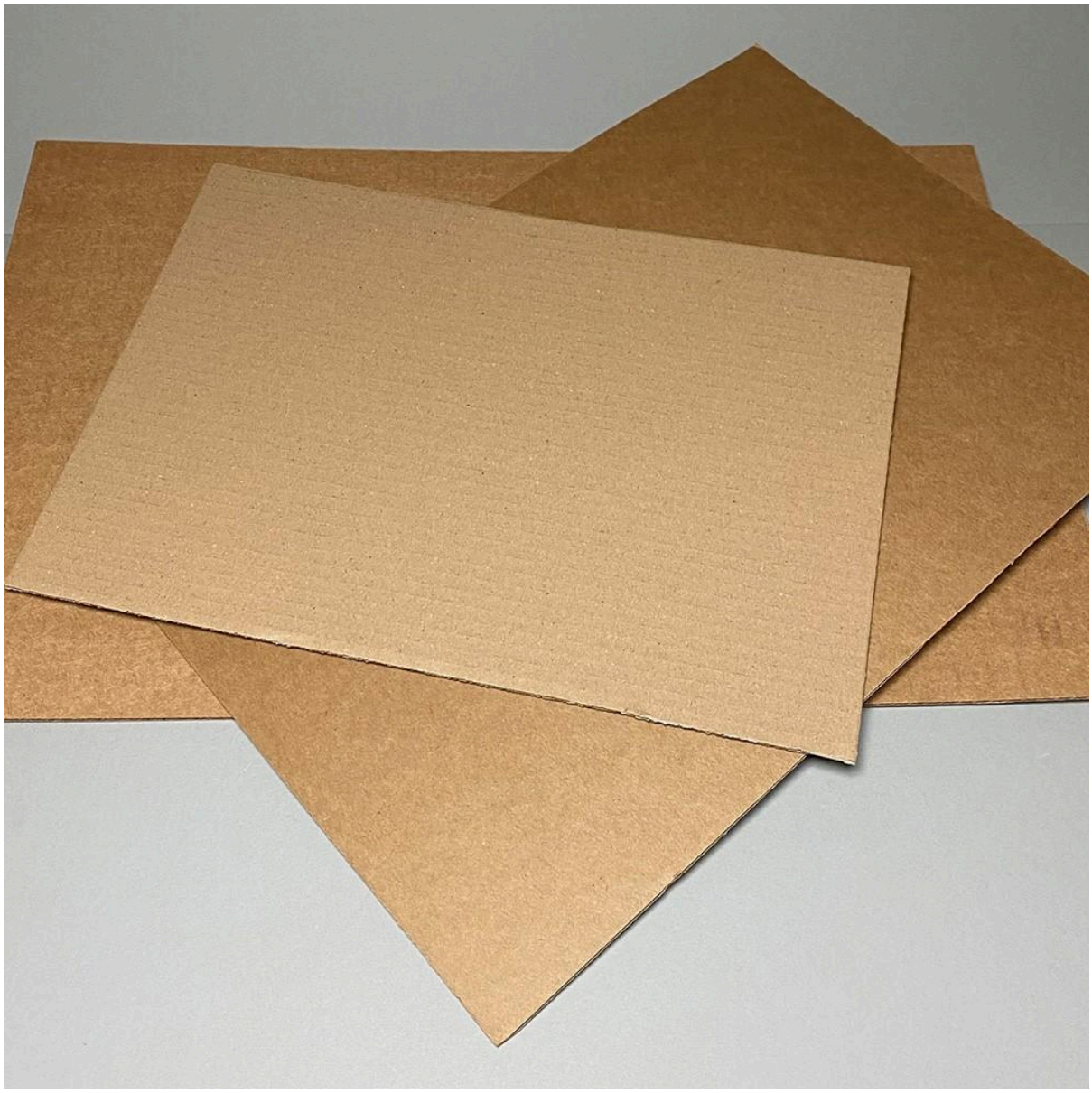
Connects buttons to Arduino pins and GND.

IV.**Foam tape**

Helps partition between two MDF sheets and facilitates the pressing of the Button Pads

V. **MDF sheets (x9)**

VI. **Cardboard (x4)**



VII. **Plywood (x1)**

Code:
https://docs.google.com/document/d/1EPtZlh2yJXB6Z0fF2O6MXjmYCcDa2gu3Gc6wTe0ueWE/edit?usp=sharing

Results: